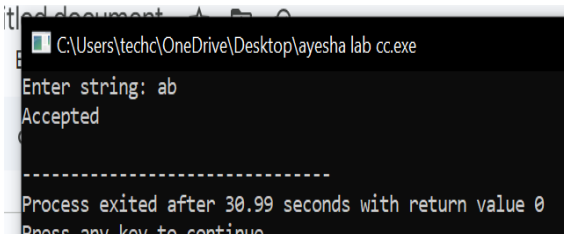
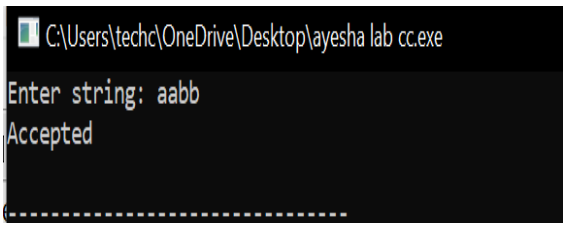
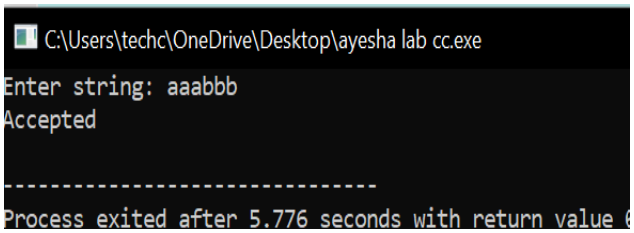
Name: Syeda Ayesha Fatima
Sap Id: 53953
Submitted to: Ma’am Hasnat

Task 1:

```
#include <iostream>
#include <string>
using namespace std;
string input;
int i = 0;
void S() {
    if(input[i] == 'a') {
        i++;        // match 'a'
        S();        // recursive call
        if(input[i] == 'b') {
            i++;    // match 'b'
        } else {
            cout << "Rejected\n";
            exit(0);
        }
    }
    // else epsilon (do nothing)
}
int main() {
    cout << "Enter string: ";
    cin >> input;
    S();
    if(i == input.length())
        cout << "Accepted\n";
    else
        cout << "Rejected\n";
}
```



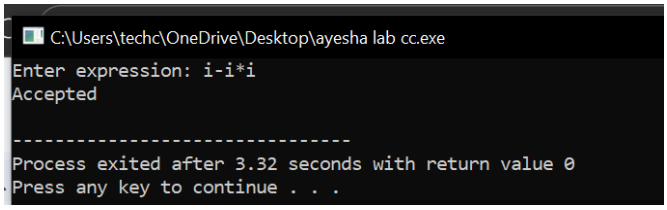




Task 2:

```
#include <iostream>
#include <string>
using namespace std;

string input;
int i = 0;

void E(); void Eprime();
void T(); void Tprime();
void F();

void match(char c) {
   if(input[i] == c)
      i++;
   else {
      cout << "Rejected\n";
      exit(0);
   }
}

void E() {
   T();
   Eprime();
}

void Eprime() {
   if(input[i] == '-') {
      match('-');
      T();
      Eprime();
   }
}

void T() {
   F();
   Tprime();
}

void Tprime() {
   if(input[i] == '*') {
      match('*');
      F();
```

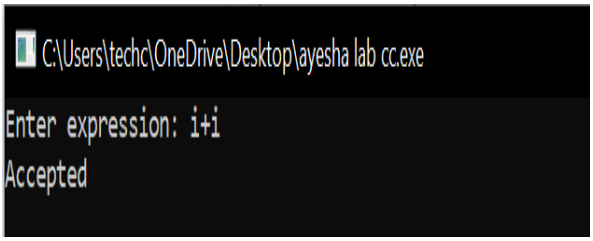
```
        Tprime();
    }
}

void F() {
    if(input[i] == 'i') {
        match('i');
    }
    else if(input[i] == '(') {
        match('(');
        E();
        match(')');
    }
    else {
        cout << "Rejected\n";
        exit(0);
    }
}

int main() {
    cout << "Enter expression: ";
    cin >> input;

    E();

    if(i == input.length())
        cout << "Accepted\n";
    else
        cout << "Rejected\n";
}
```

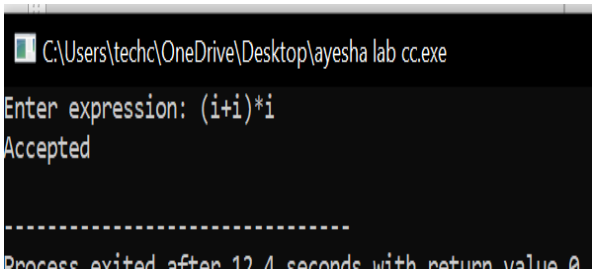


Task 3:

```
#include <iostream>
#include <string>
using namespace std;

string input;
int i = 0;

void E(); void Eprime();
void T(); void Tprime();
void F();

void match(char c) {
   if(input[i] == c)
      i++;
   else {
      cout << "Syntax Error\n";
      exit(0);
   }
}

void E() {
   T();
   Eprime();
}

void Eprime() {
   if(input[i] == '+' || input[i] == '-') {
      char op = input[i];
      match(op);
      T();
      Eprime();
   }
}

void T() {
   F();
   Tprime();
}

void Tprime() {
   if(input[i] == '*' || input[i] == '/') {
      char op = input[i];
```

```
        match(op);
        F();
        Tprime();
    }
}

void F() {
    if(input[i] == 'i') {
        match('i');
    }
    else if(input[i] == '(') {
        match('(');
        E();
        match(')');
    }
    else {
        cout << "Syntax Error\n";
        exit(0);
    }
}

int main() {
    cout << "Enter expression: ";
    cin >> input;

    E();

    if(i == input.length())
        cout << "Accepted\n";
    else
        cout << "Rejected\n";
}
```





```
C:\Users\techc\OneDrive\Desktop\ayesha lab cc.exe
Enter expression: i*i+i
Accepted

--------------------------------
Process exited after 5.106 seconds with return value 0
Press any key to continue . . .
```

Dry Run:

Dry Run:

Task 1: String: aaabbb

S -> aSb
-> a(aSb)b
-> a(a(aSb))b
-> a(a(aεb)b)b

-> aaabbb
Accepted

Task 2:
String : i*i+i

E -> TE'
T -> FT'
identifier ← F -> i i
multiplication ← T' -> *FT' i*i (passed)
addition ← E -> +TE' i*i+
T -> FT' i*i+i
T' -> ε
ε' -> ε